Mannan Shukla

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TECHNICAL SKILLS

Languages: Python, Go, Bash, TypeScript
Inference: vLLM, SGLang, TensorRT, NVIDIA Dynamo, CUDA, NCCL, PagedAttention, KV cache, continuous batching
ML & Agents: PyTorch, Hugging Face, LoRA, LangGraph, MCP, Kubeflow
Infrastructure: Kubernetes, Docker, Helm, Terraform, AWS, ArgoCD, Cilium, Slurm
Production: Postgres (RDS), Caddy, Stripe, Auth0, Prometheus, Grafana, GitHub Actions, CI/CD, GitOps

EXPERIENCE

Founder & Engineer
spawnpoint • San Francisco, CA — July 2026 - Present

- Building **spawnpoint** (getspawnpoint.com): where a team's AI lives; agents ship tools, MCP servers, AI apps
- Built the **MCP** server in **Go** (30 tools) and tuned its text until **Qwen 3.5 4B** went from 0.27 to 0.74 on evals
- Built **OAuth 2.1** (**PKCE**, rotation) and a **data plane** gate: domain-wide sharing, OAuth MCP apps, scale-to-zero
- Built **teams**: domain join, roles, **Stripe** billing, a team domain, a shared **S3** space, skills served over **MCP**
- Built an **OpenAI**/**Anthropic**-compatible **inference gateway** for open models: per-app keys, usage metering
- Replaced SSH pushes with a **pull agent** on each VM (deploys 93 s to 72 s); static sites serve from **S3**, no VM
- Run production on **AWS** (EC2, **RDS**, S3) with per-app **Postgres** and **S3**, zero-downtime deploys over **SSM**

Software Engineer
Voltage Park / Lightning AI • San Francisco, CA & Redmond, WA — July 2025 - July 2026

- Shipped hot model-weight reloading for **vLLM** inference deployments via a **Go** sidecar, cutting fine-tuned model rollout time from 10-30 minutes to seconds without restarting GPU workers
- Built core pieces of an **S3**-compatible object store (**VAST**-backed): a health-aware **P2C** client-side load balancer over CNode IPs with lock-free in-flight tracking and automatic failover, plus **Terraform**/**Vault**-managed **RDS** infra
- Built distributed inference orchestration in **Go** on **Kubernetes**, implementing deployment-target controllers and **DynamoDB** event sourcing to reconcile and recover fleets of GPU-serving replicas
- Created the **vp-cloud-sdk Python** SDK for Voltage Park's orchestration platform, exposing fleet/node/remediation APIs and a full staging release pipeline with **GitHub Actions**
- Own a multi-tenant GPU platform spanning **ArgoCD**, **Terraform**, **Helm**, **Cilium**/**Envoy**, NVIDIA GPU Operator, and **Prometheus**/**Grafana**, with tenant isolation and production observability
- Built a **Slurm** GPU health-check suite for distributed training (ECC, PCIe AER, **NCCL** stalls, **DCGM** checks, thermal) that auto-drains unhealthy nodes, creates RMA tickets, and sends **Slack** alerts to protect multi-node jobs
- Operate production **H100** clusters, resolving incidents across GPU scheduling, distributed networking, and **NCCL** workloads

Undergraduate Researcher: AIOS
Rutgers University • New Brunswick, NJ — Apr 2024 - Apr 2025

- Built an **LLM** agent recommender hitting 83% accuracy vs. **GPT-4** with a 1B model via **LoRA** fine-tuning (**Unsloth**)

Partnerships Intern
Nas.io • New York, NY — Feb 2025 - May 2025

- Built an **Instagram scraper** and **Python**/**Klaviyo** pipeline that lifted outreach 50x; led partnerships to $2K GMV

PROJECTS

BYOA: Bring Your Own Agent
- Built a **Kubernetes** inference platform that deploys agents on arbitrary **Hugging Face** models behind OpenAI-compatible APIs, using **NVIDIA Dynamo** for disaggregated serving, KV-cache-aware routing, and GPU autoscaling

TaskCLI: SWE-bench-style evaluation runner
- Built a **Python** agent-evaluation harness for SWE-bench-style task bundles with reproducible **Docker** sandboxes, hidden-test patch grading, **JUnit XML** parsing, **SQLite** artifact tracking, and OpenAI-compatible solvers

tochi.one: AI property management agents
- Built payment, lease-generation, and tenant-communication workflows in **LangGraph**/**MCP** with **Stripe** and **Twilio**

AdBlockIRL: AR ad-blocking
- Trained **YOLOv8** on **Roboflow** to detect 100+ concurrent ads at 5ms latency, applying real-time mosaic censoring via **OpenCV**; won $2000 at Bain Capital Ventures AI Hackathon

EDUCATION

Rutgers University – New Brunswick
Bachelor of Science - Computer Science • Cum Laude • Linux User Group (President) — May 2025